Project Part 3: Prototype Development, Live Demonstration & Interactive Oral Assessment

Total Marks: 25
Assessment Type: Group Project with Individual Oral Assessment
Duration: Approximately **15 minutes** per group
Assessment Schedule: To be announced

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1. Overview

In Part C, your group is required to develop a **working prototype** that demonstrates important aspects of your proposed solution. The prototype should reflect the business requirements and architectural decisions developed in Parts A and B.

You are **not expected** to implement the complete architecture or develop a production-ready application. Instead, your prototype should demonstrate sufficient functionality to help the client understand how your proposed system would look, operate, and address their requirements.

Your group will present its prototype and architecture through a live demonstration followed by an Interactive Oral (IO) assessment.

During the IO, each student will be asked questions to demonstrate their understanding of the project, architectural decisions, technical implementation, and individual contribution.

2. Project Requirements

A. Working Prototype

Your group should develop a prototype that:

- Demonstrates the main features of your proposed application.
- Provides a basic but functional user experience.
- The requirements identified in project Part A and B.
- Demonstrates some of the **technical components** or services included in your architecture.
- Usage of appropriate programming languages, frameworks, and technologies during the demo

You may use dummy or sample data where appropriate.

Important: You are not required to implement every feature or architectural component proposed in Parts A and B. However, you should be able to explain which components have been implemented and which remain part of the proposed architecture.

B. Cloud Deployment

Your group is **expected to deploy its working prototype on a suitable cloud-hosting platform**, such as AWS, Azure, Google Cloud, or another appropriate cloud-hosting service.

The objective is to demonstrate that you can make your prototype accessible online. You are **not required to deploy your complete proposed architecture** or configure complex cloud infrastructure.

Minimum Cloud Deployment Requirement: Your prototype should be accessible through a working URL hosted on a suitable cloud platform. A simple static or frontend-only deployment is acceptable

For example:

- **Static website (HTML, CSS, JavaScript):** Use a straightforward hosting service such as AWS S3 static website hosting or Azure Static Web Apps.
- **Frontend application:** Deploy using a hosting service compatible with your framework.
- **Application with a backend:** Deploy a minimal working version using an appropriate cloud service. You do not need to implement every proposed backend component.

Hint: Keep your deployment simple. If your architecture includes several components, you may implement and deploy only a small part of it. For example, you could deploy a working frontend with a few functional features and use sample data for features that are not yet implemented.

Important: ***You are not expected to purchase paid cloud services or software***. Use suitable free-tier or free-hosting options where possible, and check their current usage limits and billing conditions.

C. Architecture and Technical Implementation

Your group should be prepared to explain:

- Your final system architecture and its main components.
- How the prototype relates to the architecture developed in Parts A and B.
- The reasons behind your architectural and technology choices.
- How the frontend, backend, database, and external services communicate, where applicable.
- Relevant security, scalability, availability, or integration considerations.
- The main limitations of your prototype.
- How and where the prototype has been deployed.

D. Project Management and Collaboration

Your group should maintain evidence of development and collaboration using:

- **GitHub/GitLab:** Source code repository and development history.
- **Jira or an equivalent tool:** Project tasks, progress, and management activities.

- **Individual contribution breakdown:** A brief description of each group member's responsibilities and completed work.

These materials may be reviewed during the assessment to support questions about project development and individual contributions.

3. Live Demonstration and Interactive Oral Assessment

Each group will attend a scheduled assessment session lasting approximately **15 minutes**.

Part 1 — Group Demonstration (Approximately 5 Minutes)

Your group should focus on:

1. **Application overview:** Briefly describe the client's problem and your proposed solution.
2. **Working prototype:** Demonstrate the main functional features using your deployed application.
3. **Architecture and deployment:** Explain the final architecture diagram, key technologies, and how the prototype has been deployed.

Your GitHub/GitLab repository, Jira dashboard, and other project evidence should be available for inspection if requested by the examiner. You do not need to present these in detail unless asked.

All group members must attend. Equal speaking time during the group demonstration is not required. Individual understanding will be assessed separately.

Part 2 — Individual Oral Assessment (Approximately 10 Minutes per Group)

Following the demonstration, the examiner will ask individual questions to group members.

Each student will be asked **one or more questions**, with possible follow-up questions, depending on the project and the student's responses.

Questions may cover:

- Individual contributions and responsibilities.
- How specific project features or components work.
- Architectural decisions and technology choices.
- Communication between different system components.
- Cloud deployment and implementation decisions.
- Security, scalability, availability, databases, or integration, where relevant.
- Differences between the proposed architecture and the working prototype.
- Development challenges and how they were addressed.

Students may refer to their architecture diagram, prototype, source code, or other project evidence when explaining their answers.

Individual assessment is based on understanding and evidence of contribution, not on how much a student speaks during the group demonstration.

4. Marking Rubric — 25 Marks

Assessment Criterion	Marks	Assessment Type
1. Working Prototype, Cloud Deployment & Demonstration	**10**	Group
2. Architecture & Technical Implementation	**7**	Group
3. Individual Technical Understanding	**5**	Individual
4. Individual Contribution & Project Ownership	**3**	Individual
Total	**25**	

Criterion 1 — Working Prototype, Cloud Deployment & Demonstration (10 Marks)

Assessment considers:

- Functionality of the implemented features.
- Relevance to the client's main requirements.
- Usability and clarity of the prototype.
- Evidence of successful cloud deployment.
- Quality and effectiveness of the live demonstration.

Criterion 2 — Architecture & Technical Implementation (7 Marks)

Assessment considers:

- Clarity and correctness of the architecture diagram.
- Consistency between the proposed architecture and implemented prototype.
- Justification of architecture and technology choices.
- Appropriate use of technical components and services.
- Understanding of implementation limitations and relevant design considerations.

Criterion 3 — Individual Technical Understanding (5 Marks)

Assessment considers:

- Understanding of relevant technical and architectural decisions.
- Ability to explain how components work together.
- Accuracy and clarity of answers.
- Ability to justify relevant technology choices.

Criterion 4 — Individual Contribution & Project Ownership (3 Marks)

Assessment considers:

- Ability to explain work personally undertaken.
- Evidence of participation and contribution.
- Understanding of the features, components, or tasks for which the student was responsible.

Evidence may include source code, repository activity, project management tasks, documentation, architecture design, testing, or other relevant project work.

5. Submission Requirements

Each group must submit **one PDF document** through Loop containing:

1. Group name and student names/IDs.
2. Link to the cloud-deployed working prototype.
3. Link to the GitHub/GitLab repository.
4. Link to the Jira/project management dashboard.
5. Final architecture diagram or a link to it.
6. Breakdown of individual responsibilities and contributions.
7. Any additional links relevant to the assessment.

Filename: `CSC1057_PartC_GroupName.pdf`

Access Requirements:

- All submitted links must be accessible to the assessment team without additional permission requests.
- If your application requires authentication, provide suitable demonstration credentials using non-sensitive test data.
- Private repositories may remain private, provided the examiners have been granted the necessary access in advance.
- Check that all links work before submitting.

A recorded video presentation is **not required**, as Part C will be assessed through a live demonstration and interactive oral assessment.

6. Important Guidelines

- All group members are expected to attend their scheduled assessment session.
- Every student should be prepared to answer individual questions.
- You may use appropriate programming languages, frameworks, and development tools.
- You are not required to implement your complete proposed architecture.

- Dummy or sample data may be used, provided that simulated functionality is clearly identified.
- A simple cloud-hosted deployment is sufficient; complex infrastructure is not required.
- Clearly distinguish between implemented features and proposed future features.
- Be prepared to demonstrate project evidence, including code, architecture, and project management activities.
- Marks will be based on demonstrated functionality, architecture justification, technical understanding, and individual contributions—not simply the visual appearance or complexity of the application.
- You are not expected to purchase paid services or software.